

Advanced Signal Processing

Vector Kalman Filtering

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1 Vector Kalman Filter — Target Tracking

In this problem we consider the same situation as described in Section 5.4 of the lecture notes. We encourage you to carefully read all the provided information on this example.

Assume that $U(t)$ and $V(t)$ represent, respectively, the position and the velocity of a target at time t . Then we can write

$$U(t) = \int_0^t V(\tau) d\tau + U(0).$$

Using a sampling time of one second (denoted by T_s in the lecture notes), this integral equation can be approximated by the following difference equation

$$V(n-1) = U(n) - U(n-1), \quad n \geq 1. \quad (1)$$

We assume that the sequence

$$G(n) = V(n) - V(n-1), \quad n \geq 1 \quad (2)$$

of differences between consecutive velocity samples (acceleration) can be modeled as a white Gaussian driving sequence with variance 0.0001, i.e. we assume that

$$\begin{aligned} \mathbb{E}[G(n)] &= 0, \\ \mathbb{E}[G(n)G(n+k)] &= 0.0001\delta(k). \end{aligned}$$

Now combining (1) and (2) in matrix notation yields the state model

$$\begin{bmatrix} U(n) \\ V(n) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} U(n-1) \\ V(n-1) \end{bmatrix} + \begin{bmatrix} 0 \\ G(n) \end{bmatrix}, \quad n \geq 1. \quad (3)$$

Written in a compact form, (3) becomes

$$\mathbf{Y}(n) = \mathbf{H}(n)\mathbf{Y}(n-1) + \mathbf{Z}(n), \quad n \geq 1,$$

where

$$\mathbf{Y}(n) := \begin{bmatrix} U(n) & V(n) \end{bmatrix}^\top, \quad \mathbf{H}(n) := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{Z}(n) := \begin{bmatrix} 0 & G(n) \end{bmatrix}^\top.$$

From the above assumptions on $\{G(n)\}$, the covariance matrix of $\mathbf{Z}(n)$ reads

$$\mathbf{Q}_Z(n) := \mathbb{E}[\mathbf{Z}(n)\mathbf{Z}(n)^\top] = \begin{bmatrix} 0 & 0 \\ 0 & 0.0001 \end{bmatrix}.$$

We model the observations as

$$X(n) = U(n) + W(n), \quad n \geq 1, \quad (4)$$

where $X(n)$ is a scalar, representing the observed position of the target and $\{W(n)\}$ is a white Gaussian noise sequence with variance $\sigma_W^2 = 0.1$. Written in matrix-vector notation, (4) becomes

$$X(n) = \mathbf{A}(n)\mathbf{Y}(n) + W(n),$$

where $\mathbf{A}(n) = \begin{bmatrix} 1 & 0 \end{bmatrix}$.

1. Extend the state and observation model to the 2-dimensional (2-D) case, i.e. movement in the plane.
2. State $\mathbf{H}(n)$, $\mathbf{Y}(n)$, $\mathbf{Z}(n)$, $\mathbf{Q}(n)$ and all the other matrices and vectors used in the Kalman filter. Implement the vector Kalman filter for the 2-D case in Matlab. For initialization purposes, we assume that the target starts at $x = 0$ and $y = 0$, $\hat{\mathbf{Y}}(0|0) = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}^\top$, and $\mathbf{R}(0|0) = \mathbf{I}$.

The zip-file *targetTracks.zip* available in Moodle contains four different trajectory files named *Track1.mat*, *Track2.mat*, *Track3.mat* and *Track4.mat*. Each trajectory contain two vectors forming a sequence of true 2-D coordinates of the target. The X-coordinate is in the vector named *Xdir* and the Y-coordinate is named *Ydir*.

3. Plot Track1 by using *Xdir* and *Ydir* as the x axis and y axis, respectively. Generate noisy observations for both directions.
4. Apply the Kalman filter to estimate the position of the target by using the noisy observations. Investigate the behavior of the simulated mean squared estimation error (MSE) as a function of n by running multiple simulations. Compare the obtained MSE curve with the theoretical MSE $R(n|n)$.
5. Now, we want to investigate how to tune the parameters of the Kalman filter to increase the estimation accuracy by adjusting the variance of the driving process: (1) adjust the variance of the driving process on the x-direction while keeping the variance of the driving process on the y-direction to be 0.0001. (2) adjust the variance of the driving process on the y-direction while keeping the variance of the driving process on the x-direction to be 0.0001. Compare the predicted, the estimated and the true trajectory of the target. Compare the simulated MSE when you adjust the variance of the driving process for these two cases. State your conclusions.
6. Repeat the same procedures for the remaining tracks. How do you adjust your Kalman filter parameters to increase the estimation accuracy? State your conclusions.